Convolutional Neural Networks (CNN)

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$$y[m,n] = x[m,n] * h[m,n] = \sum_j \sum_i x[i,j]h[m-i,n-j]$$

Convolution in image processing

x

4	2	9	4	7	1
3	1	5	3	2	5
8	7	1	2	6	2
6	3	0	1	8	1
1	4	1	5	7	8
2	0	9	1	3	9

*

h

1	0	-1
1	0	-1
1	0	-1

=

y

0	1	0	1
11	5	-10	-2
13	6	-19	-3
-1	0	-8	-11

2D convolution example

4	2	9	4	7	1
3	1	5	3	2	5
8	7	1	2	6	2
6	3	0	1	8	1
1	4	1	5	7	8
2	0	9	1	3	9

*

h

1	0	-1
1	0	-1
1	0	-1

=

y

0	1	0	1
11	5	-10	-2
13	6	-19	-3
-1	0	-8	-11

(4)(1)+(2)(0)+(9)(-1)+(3)(1)+(1)(0)+(5)(-1)+(8)(1)+(7)(0)+(1)(-1) =
4+0-9+3+0-5+8+0-1 = 0

x

4	2	9	4	7	1
3	1	5	3	2	5
8	7	1	2	6	2
6	3	0	1	8	1
1	4	1	5	7	8
2	0	9	1	3	9

*

h

1	0	-1
1	0	-1
1	0	-1

=

y

0	1	0	1
11	5	-10	-2
13	6	-19	-3
-1	0	-8	-11

(2)(1)+(6)(0)+(2)(-1)+(1)(1)+(8)(0)+(1)(-1)+(5)(1)+(7)(0)+(8)(-1) =
2+0-2+1+0-1+5+0-8 = -3

2D convolution example

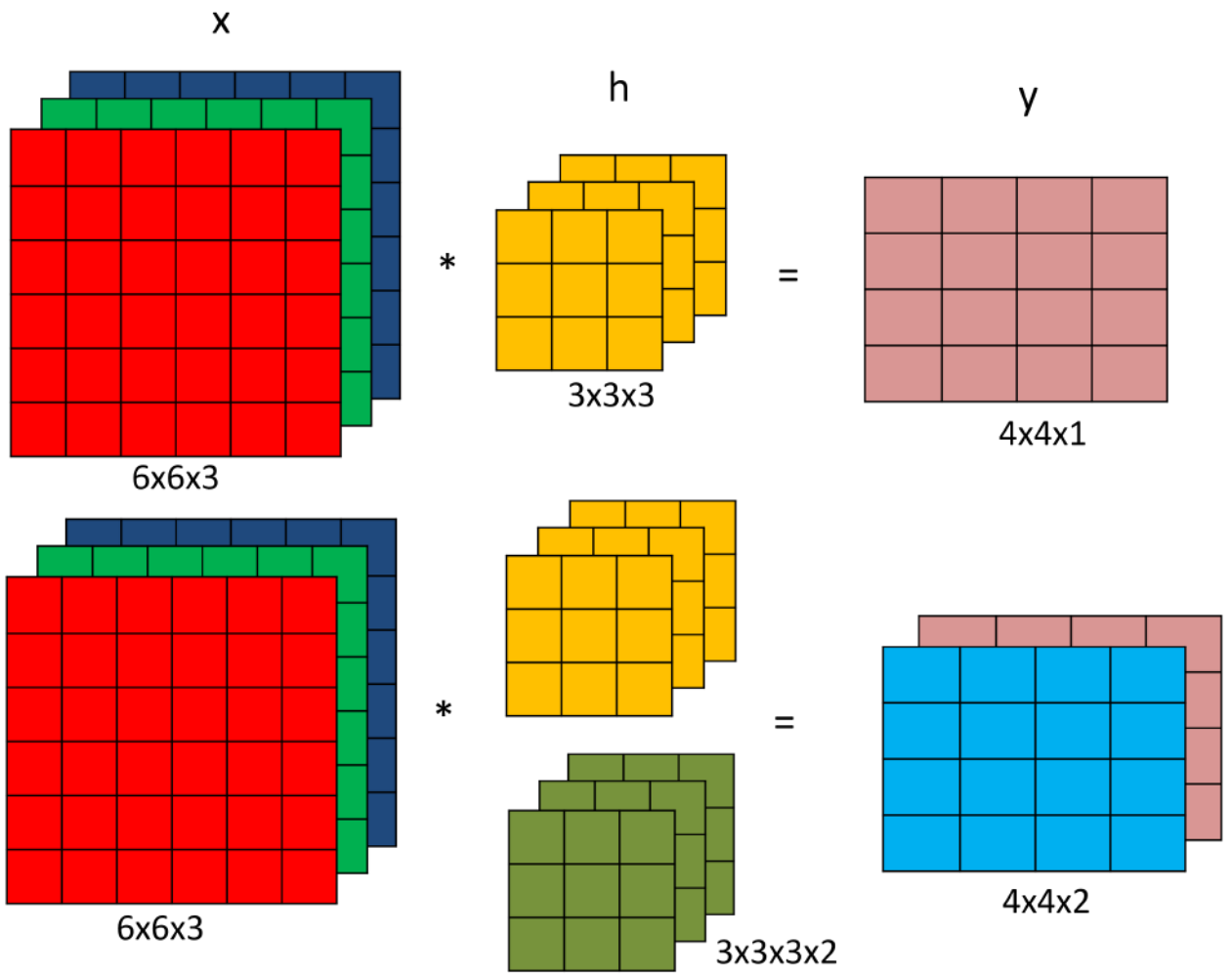


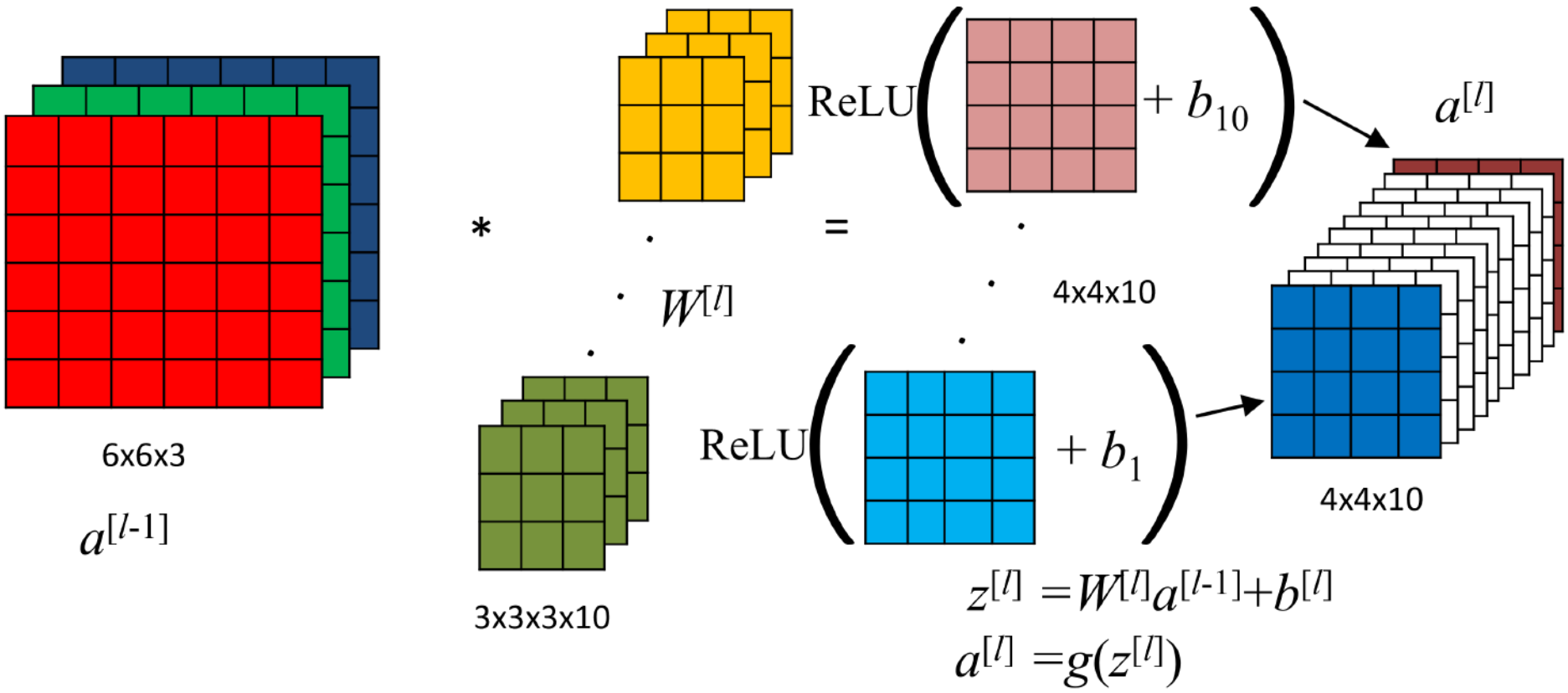


edge detection with convolution

see convolution.ipynb

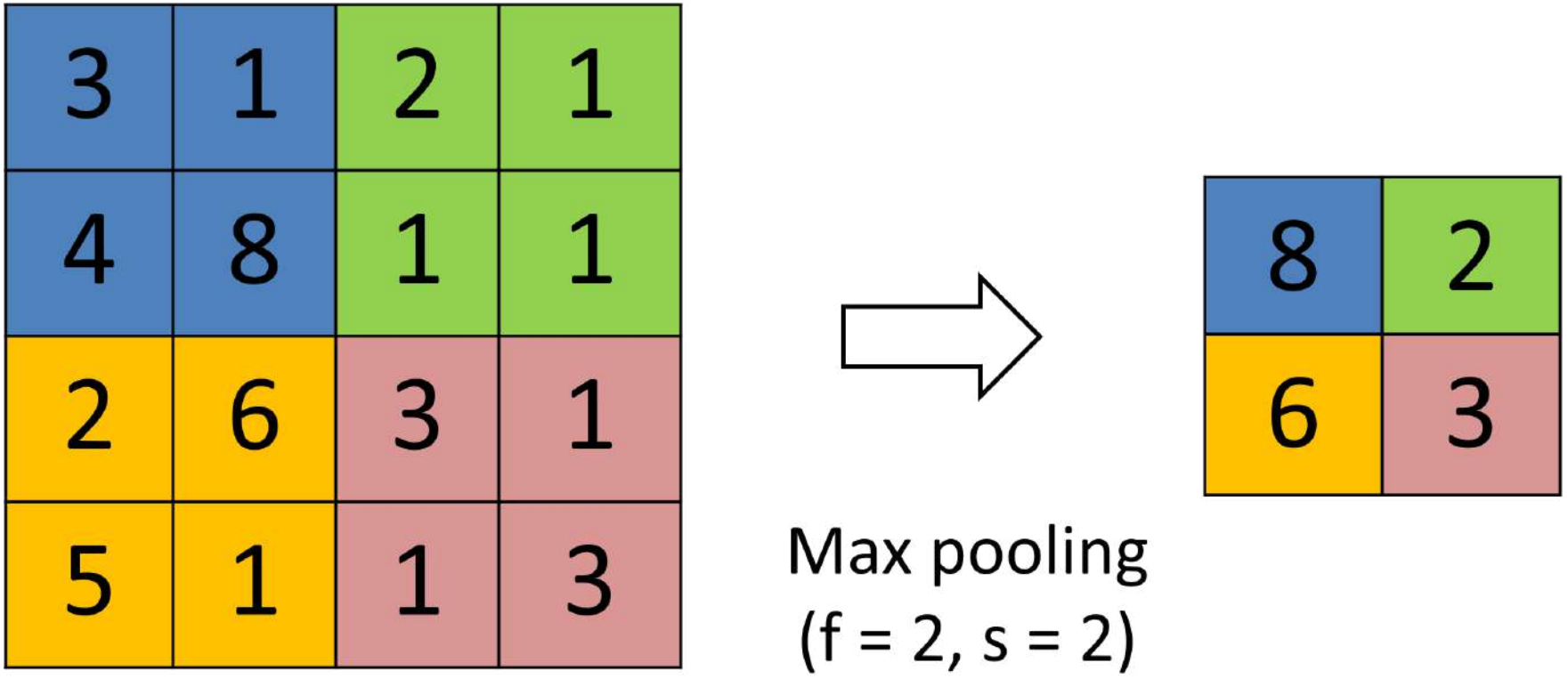
3D convolution



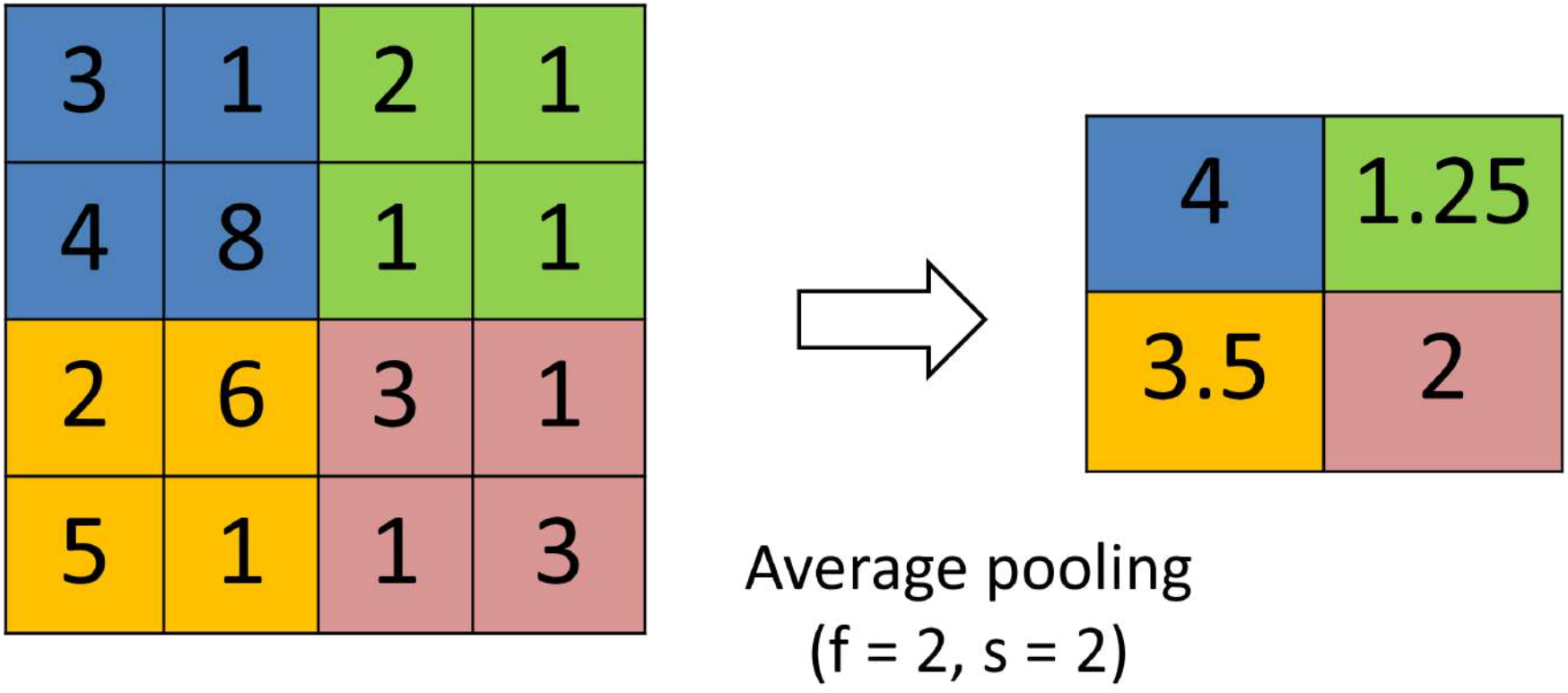


convolution layer in NN

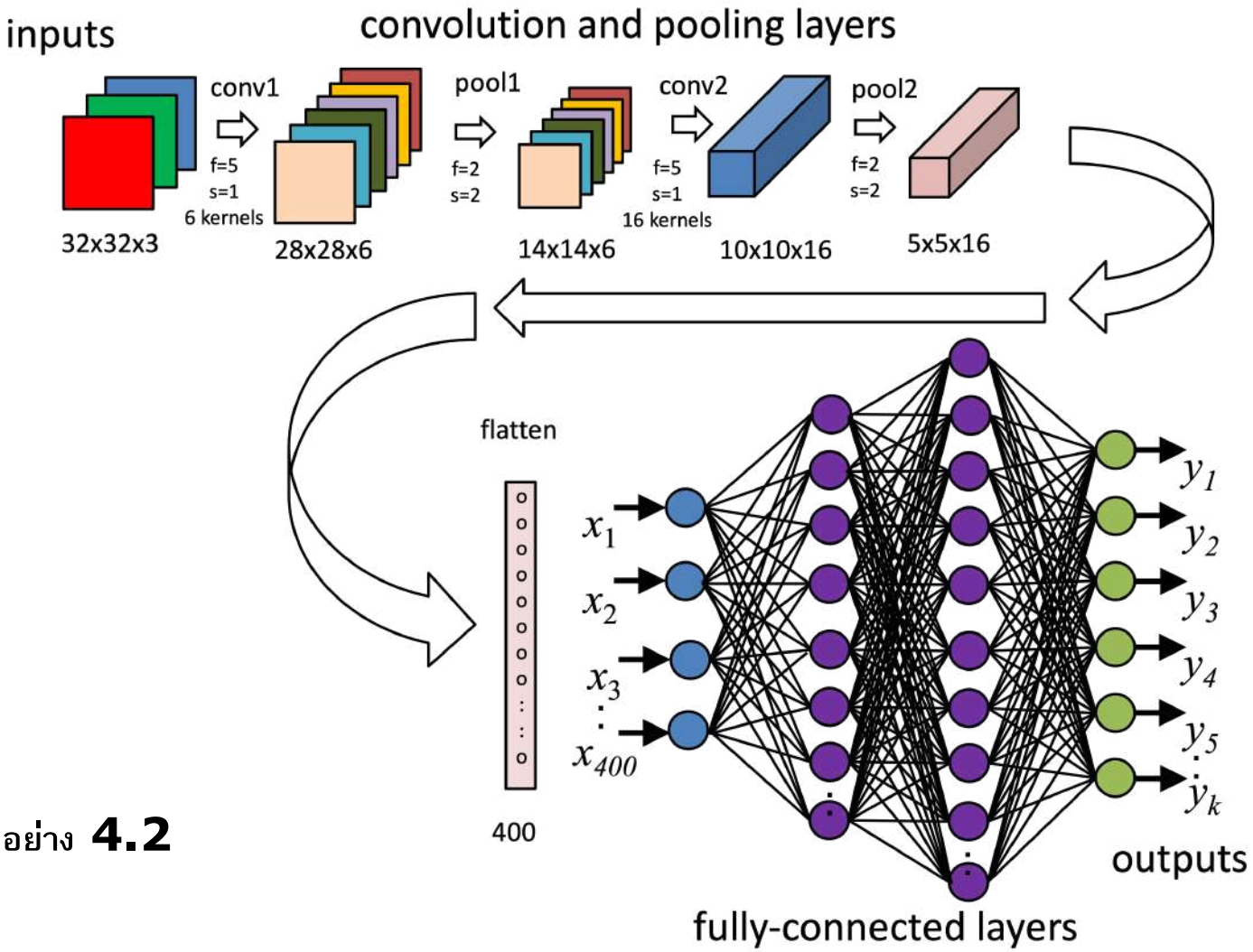
max pooling layer



average pooling layer



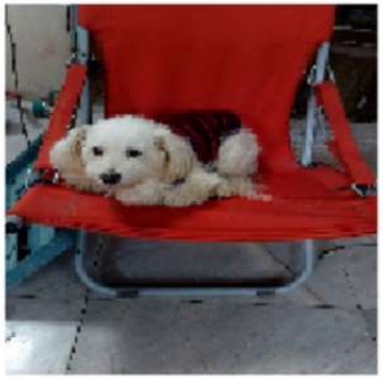
CNN model example



see ตัวอย่าง **4.2**

cat/dog classification

```
[1.]
jackie.jpg is a dog
```



```
[1.]
maam.jpeg is a dog
```



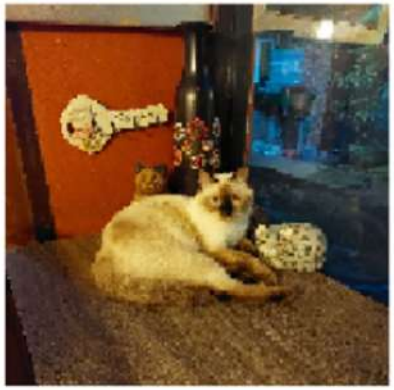
```
[0.]
dollar_scale.jpg is a cat
```



```
[1.]
fongbeer1.jpeg is a dog
```



```
[1.]
bingzoo.jpg is a dog
```



```
[0.]
ninja_bingz.jpg is a cat
```



DNN/CNN comparision (SVHN)

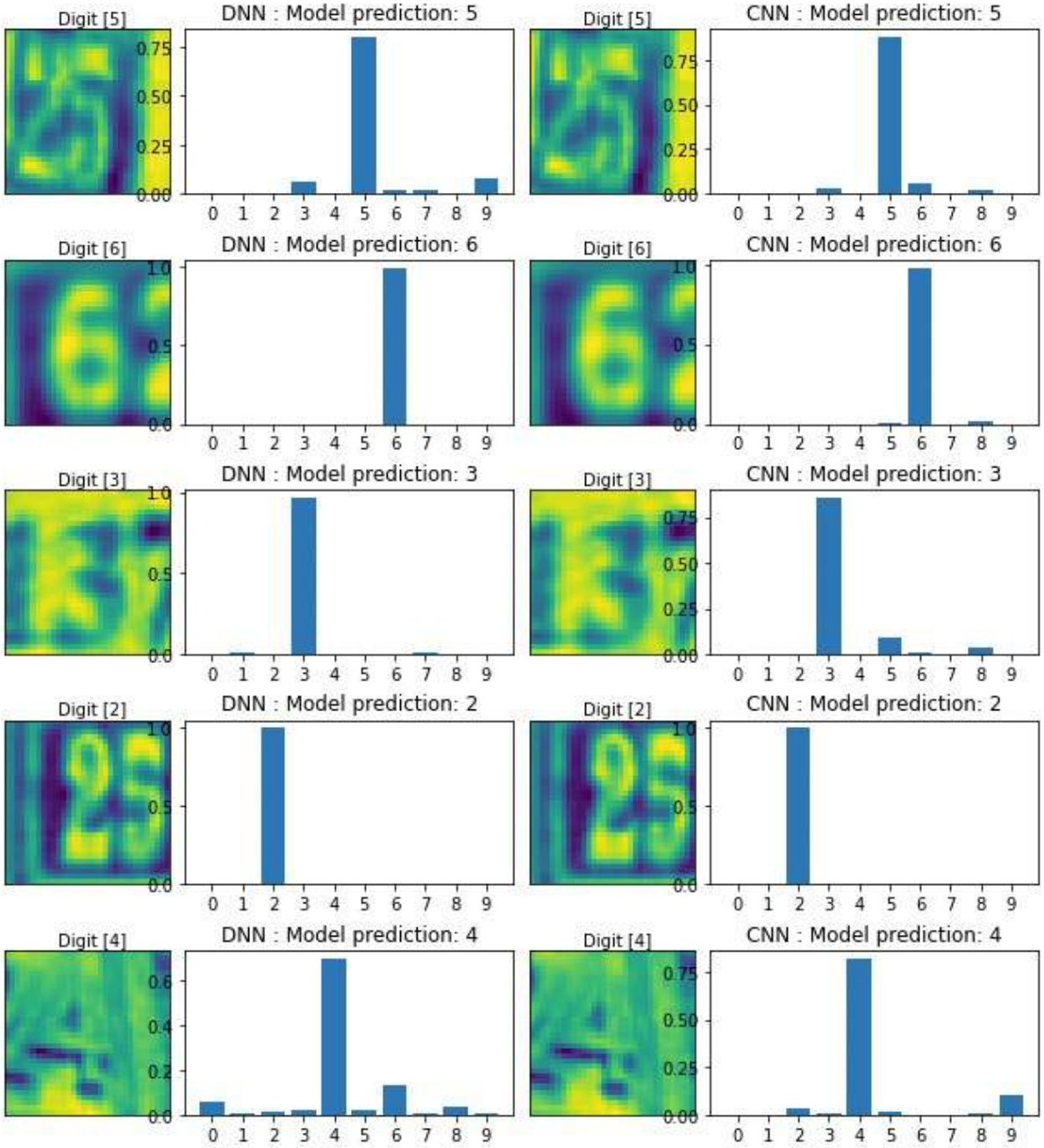


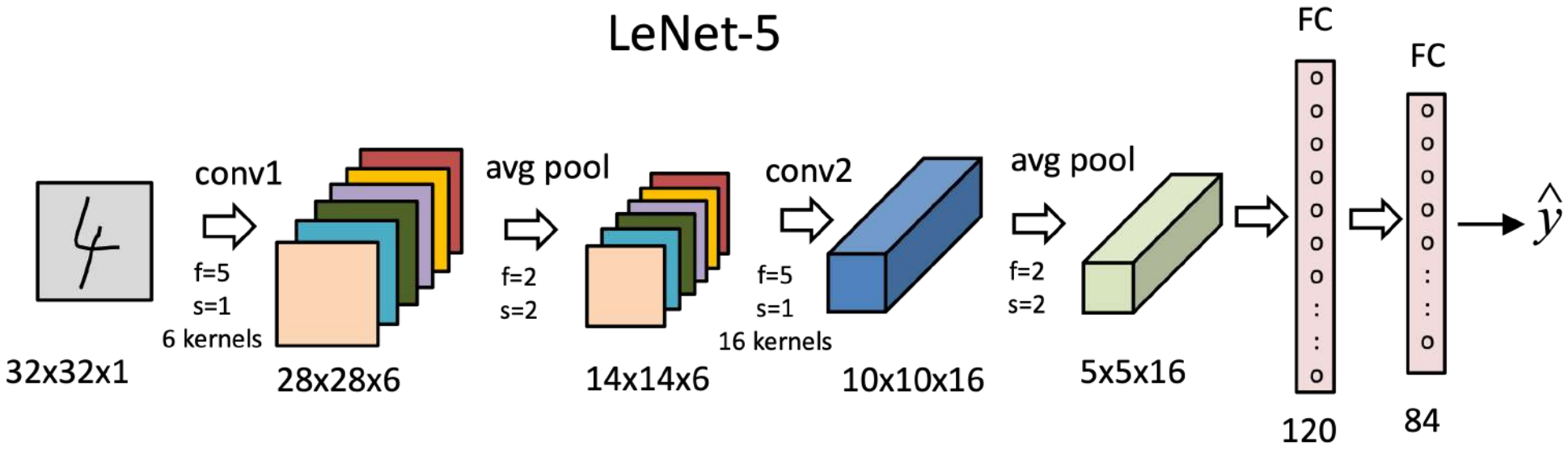
image augmentation



CNN case study

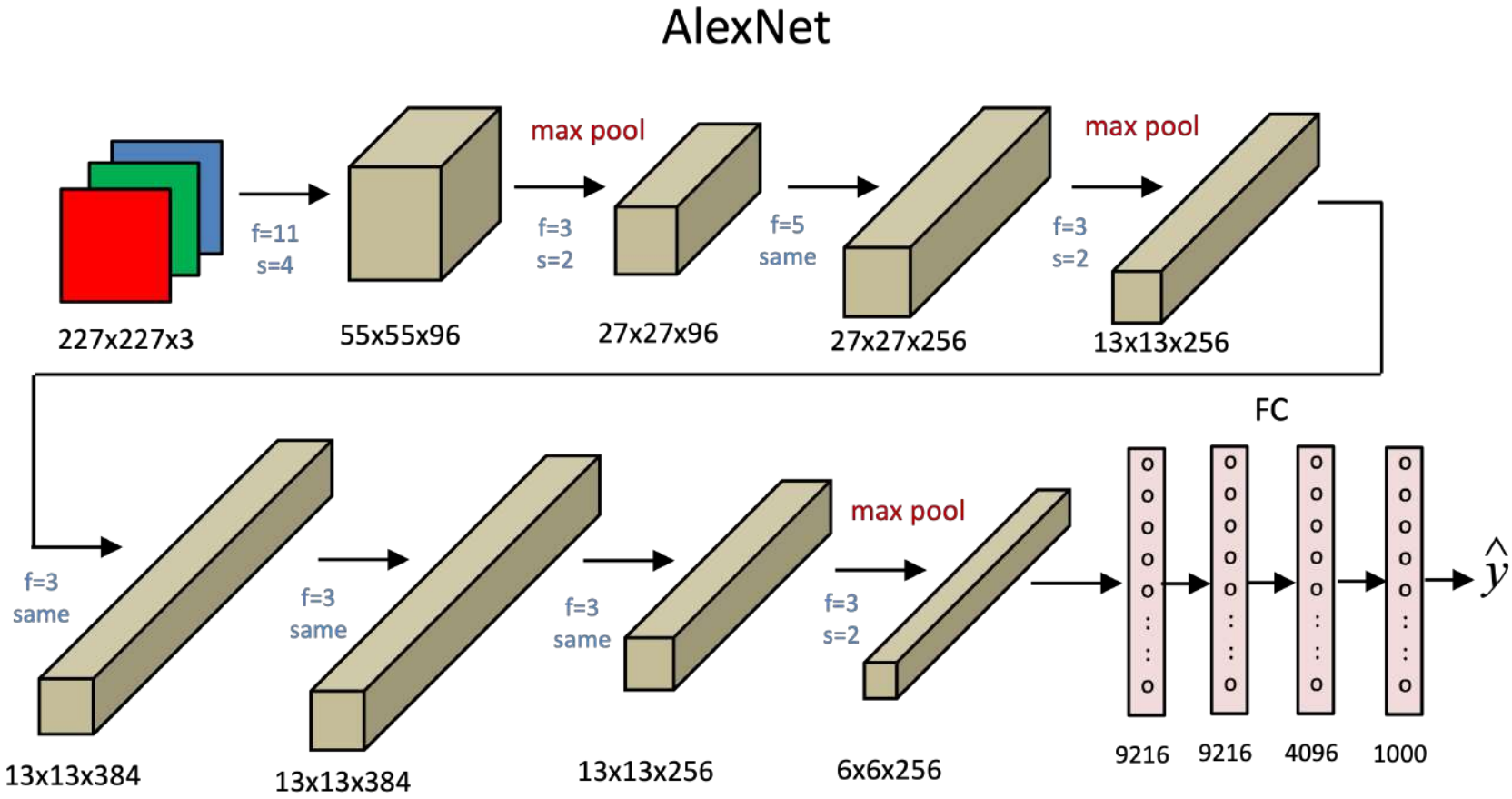
LeNet-5
AlexNet
VGG-16
ResNets
Inception model

LeNet-5



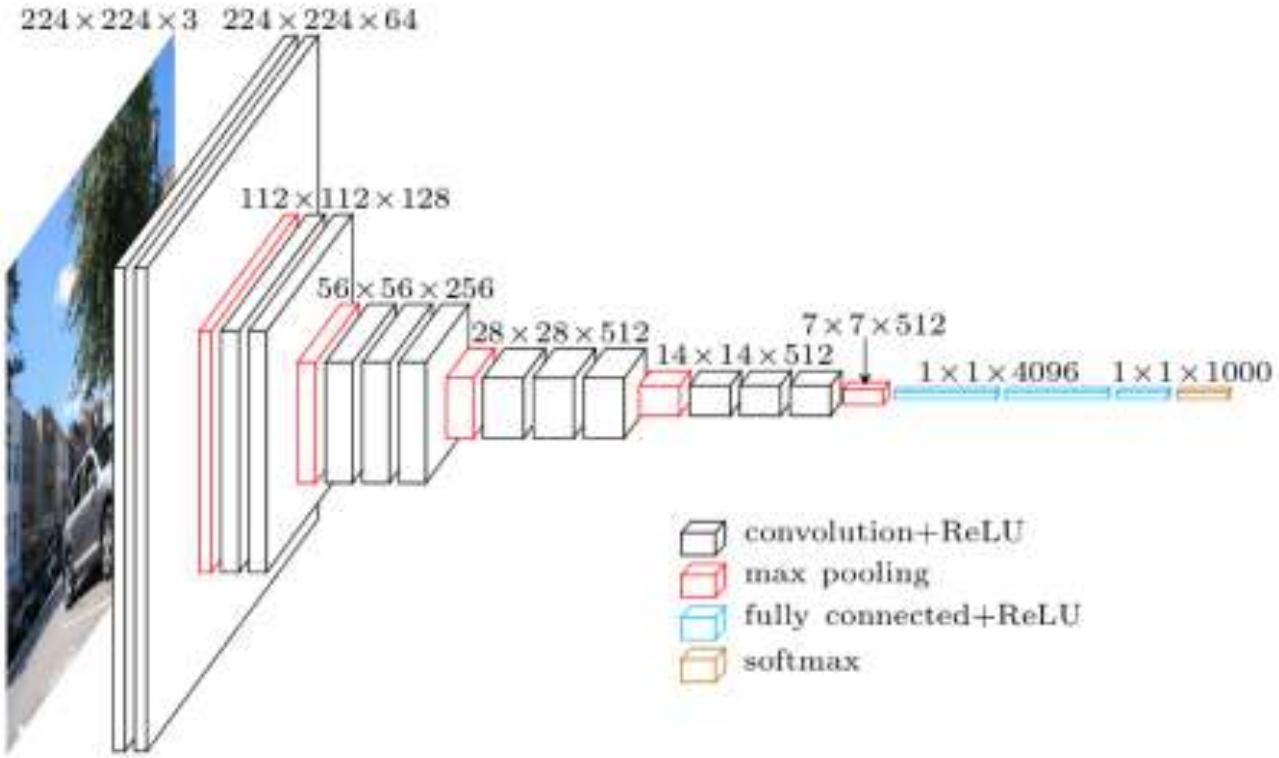
LeCun, Y.; Boser, B.; Denker, J. S.; Henderson, D.; Howard, R. E.; Hubbard, W. & Jackel, L. D. (1989). Backpropagation applied to handwritten zip code recognition. Neural Computation, 1(4):541-551.

AlexNet



Krizhevsky, Alex; Sutskever, Ilya; Hinton, Geoffrey E. (2017-05-24). "ImageNet classification with deep convolutional neural networks"

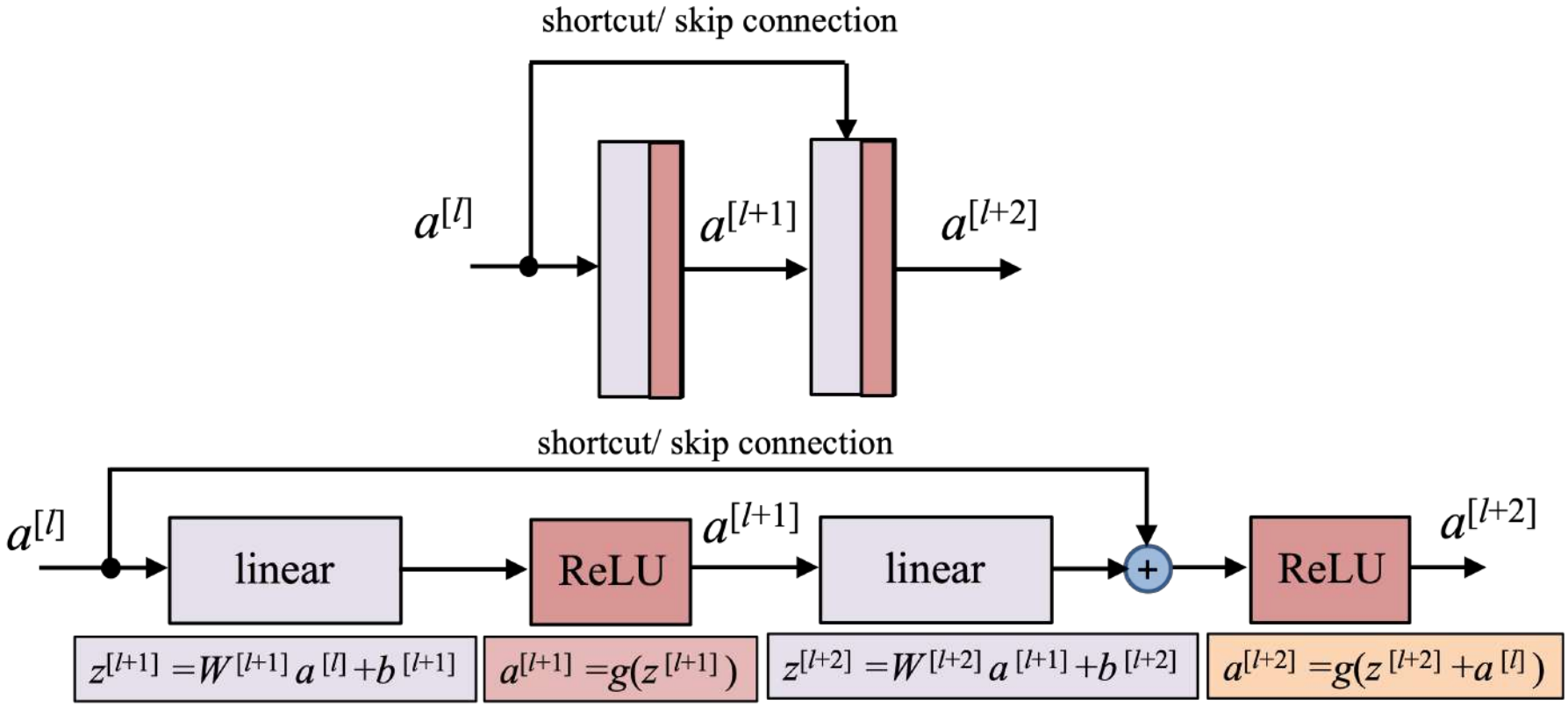
VGG-16



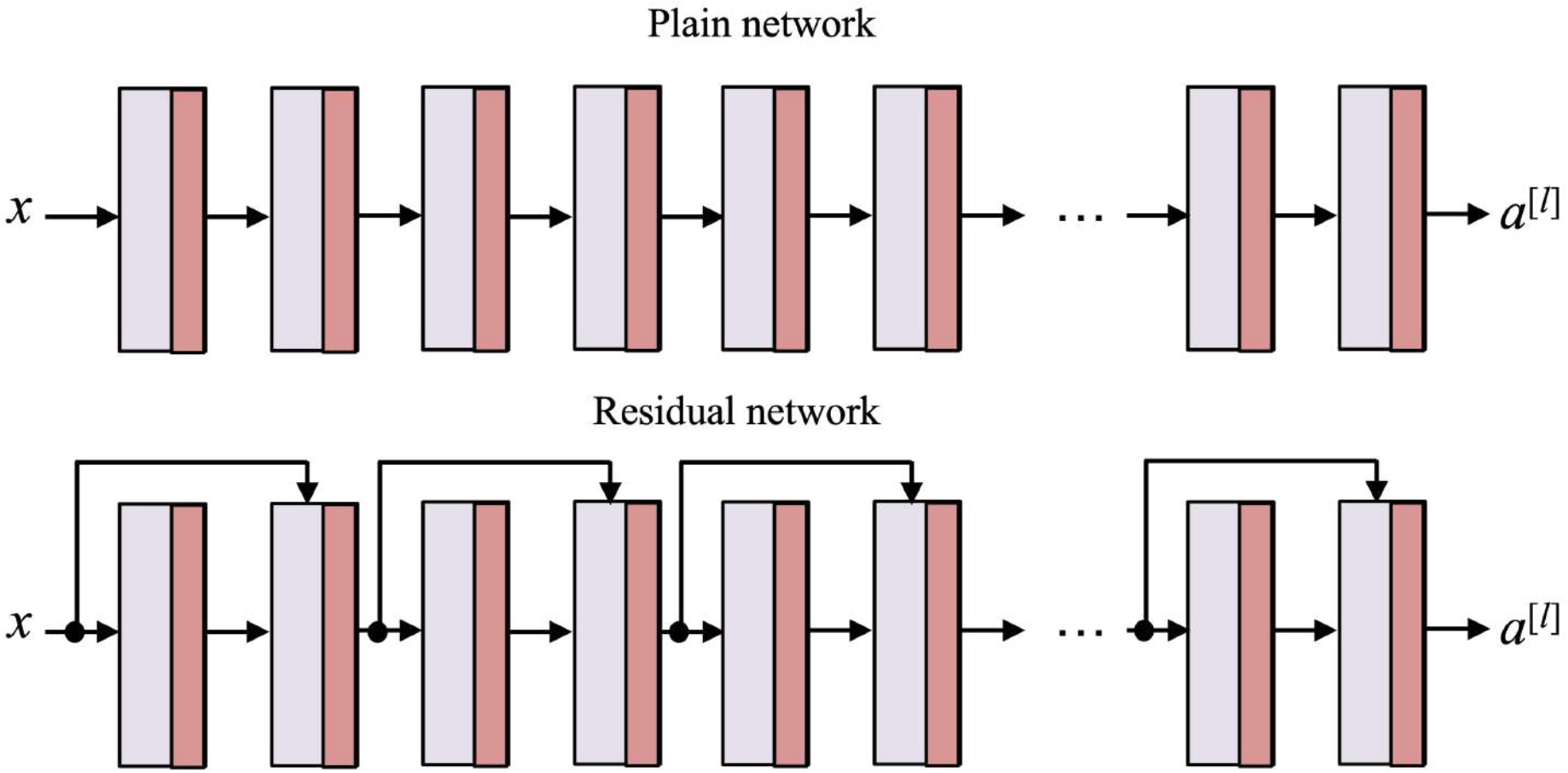
•Simonyan Sisserman. Very deep convolution networks for large-scale image recognition. 2015.

ResNets

He, K; Zhang X; Ren S.; Sun J.; Deep residual learning for image recognition. CVPR. 2015.



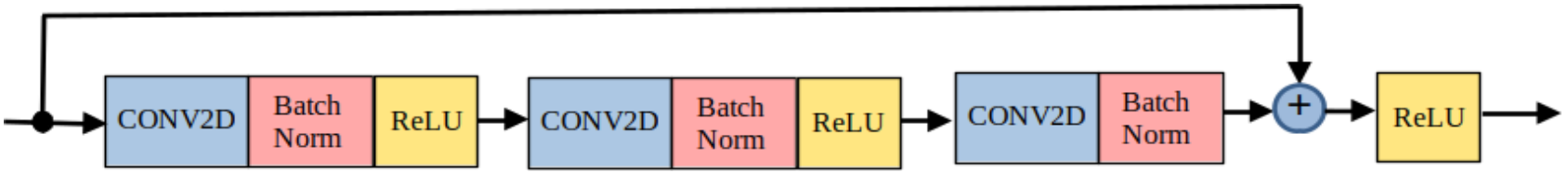
Residual block



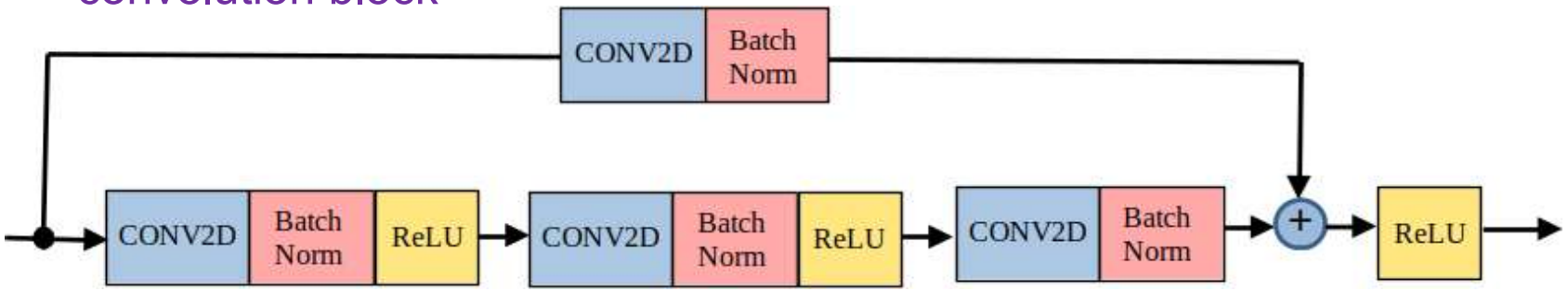
Plain v.s. residual network

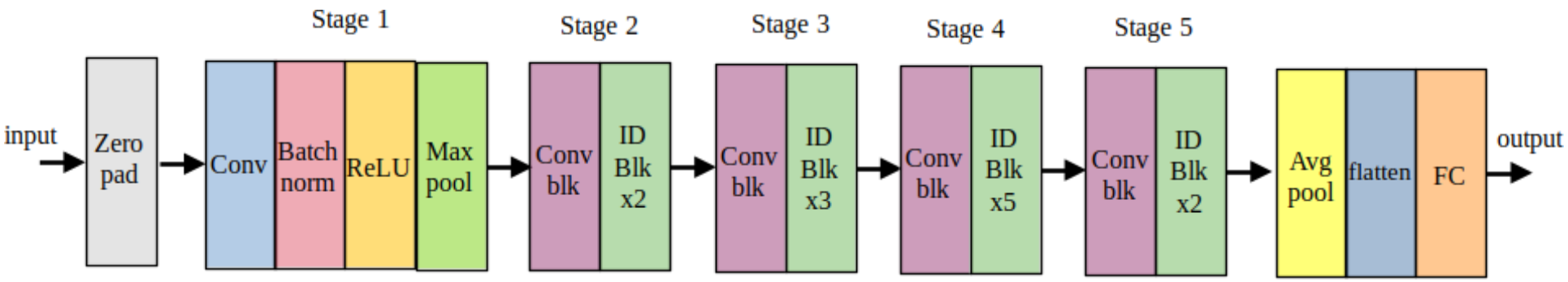
construct ResNets in TensorFlow

identity block



convolution block



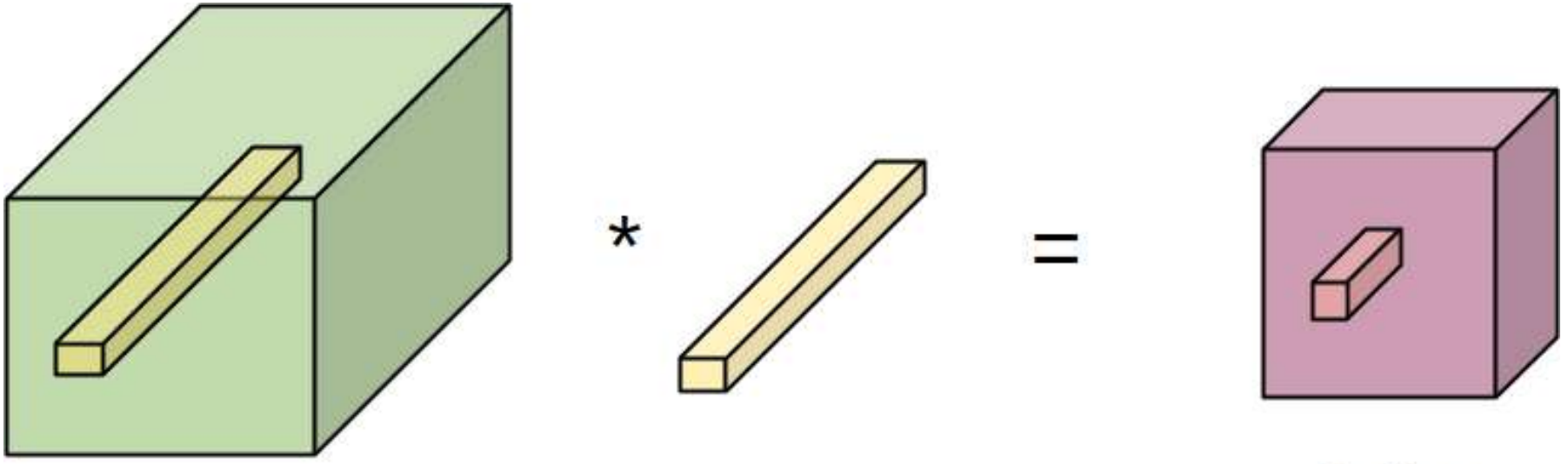


ResNet50 model

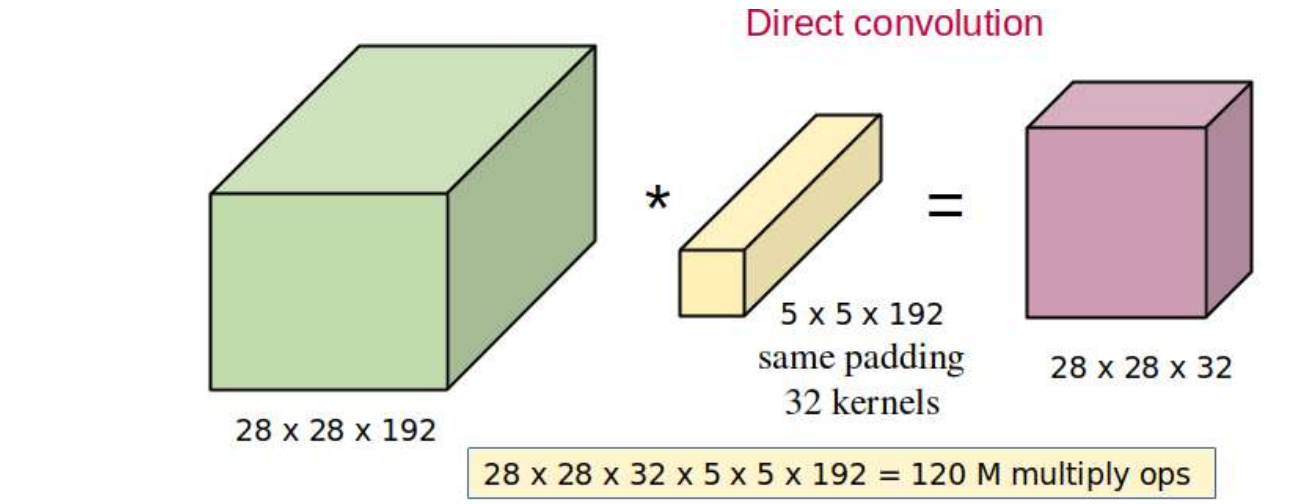
Inception model

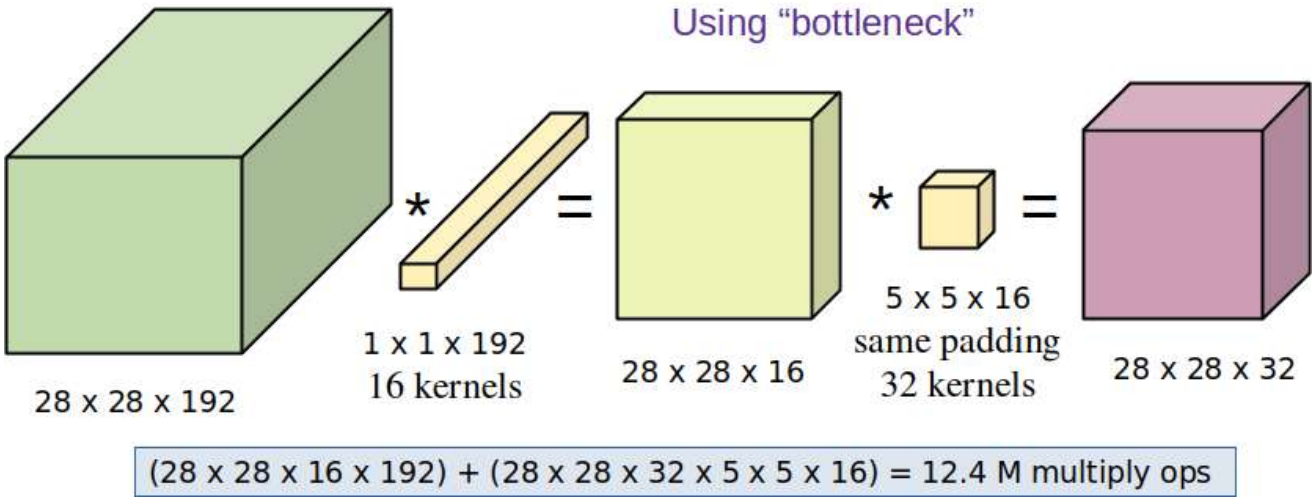
Szegedy et al. Going deeper with convolutions. 2014.

Convolution using 1x1 kernel

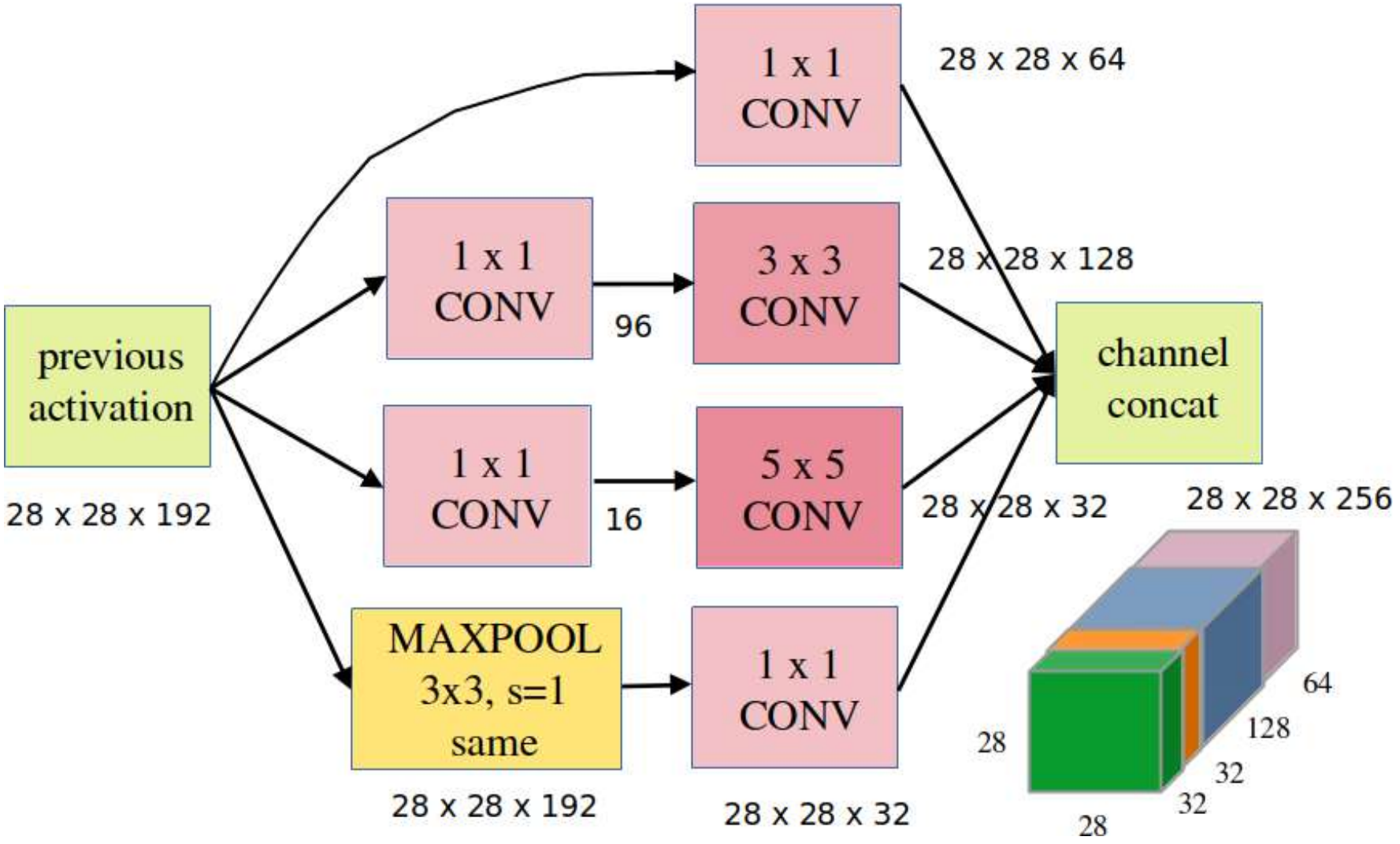


Implement “bottleneck” with 1x1 convolution

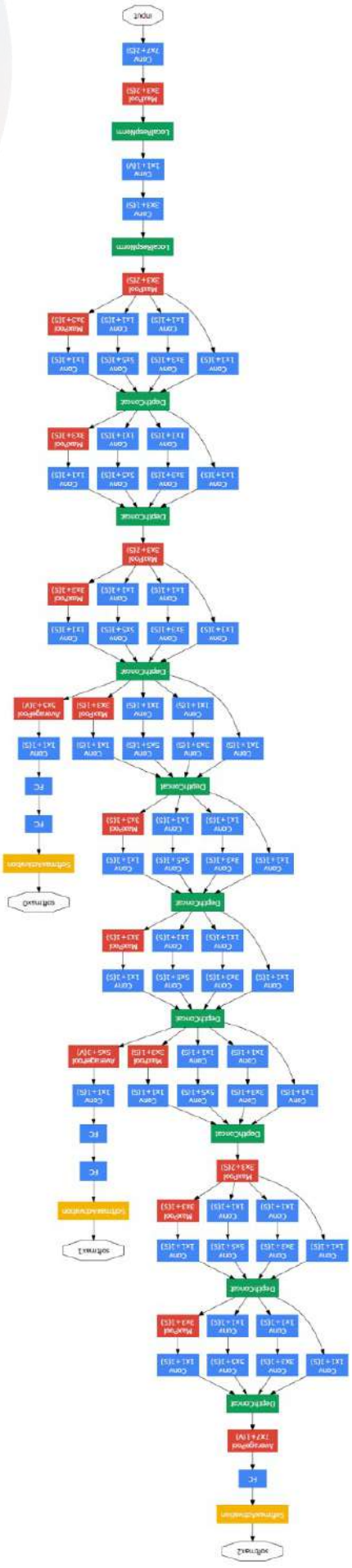


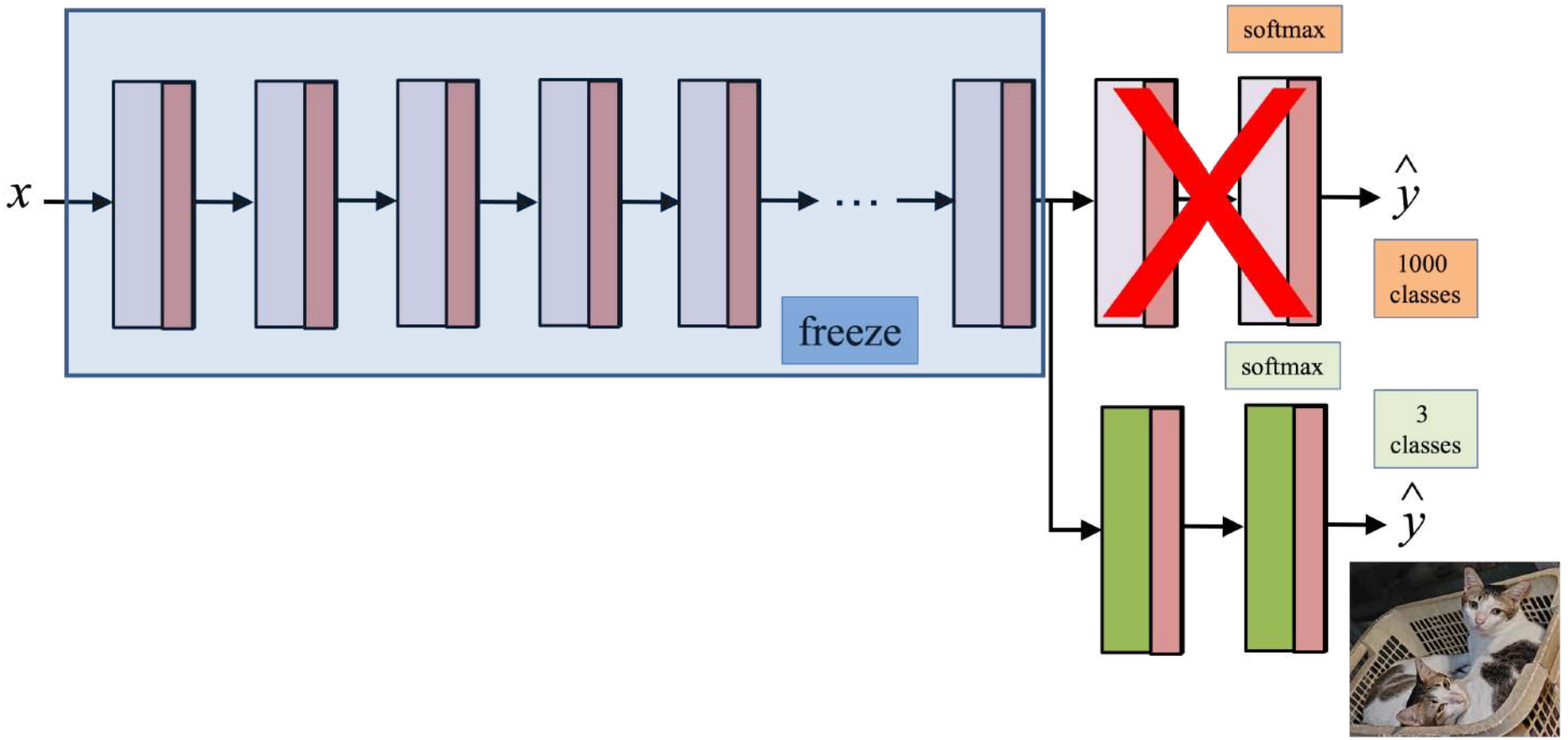


Inception module



googLenet



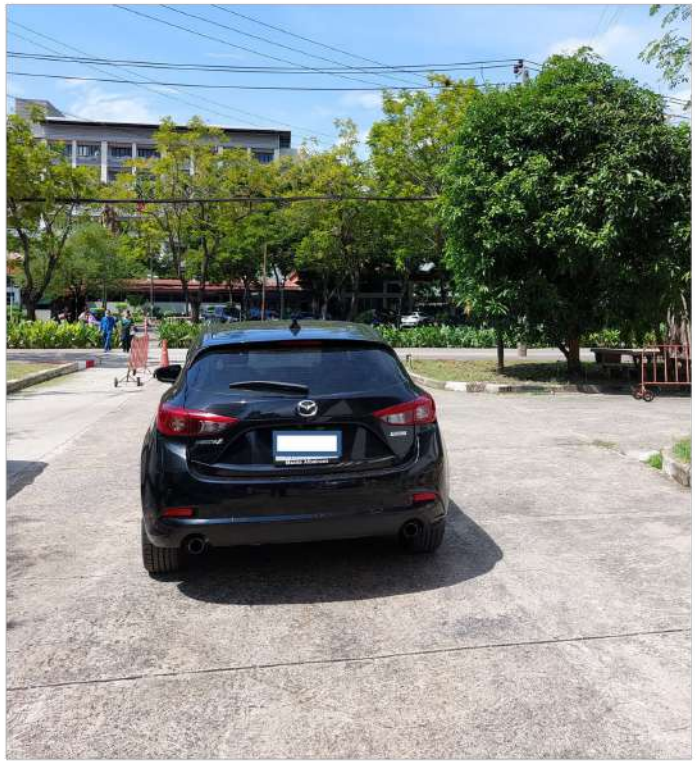


transfer learning

Object detection

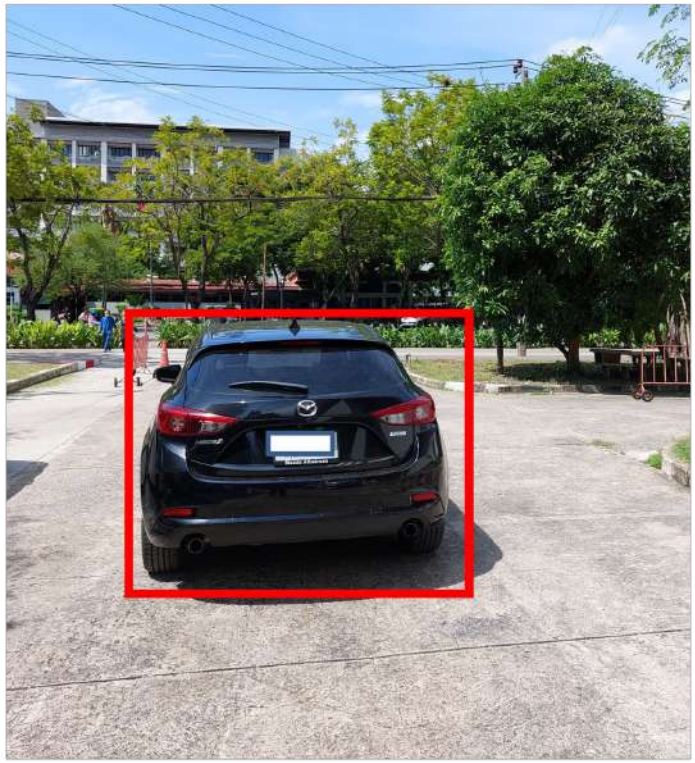
Object localization

Classification



Object : car

Classification with localization



Object : car
Position : b_x , b_y , b_w , b_h

Ex : Classify and localize the following objects

Pedestrians
Cars
Motorcycles
Background (no object 1-3)

Define output vector (label)

image input



$p_c = 1, c_1 = 0, c_2 = 1, c_3 = 0$

CNN

$$y = \begin{bmatrix} p_c \\ b_x \\ b_y \\ b_w \\ b_h \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

object detected?

box coordiates

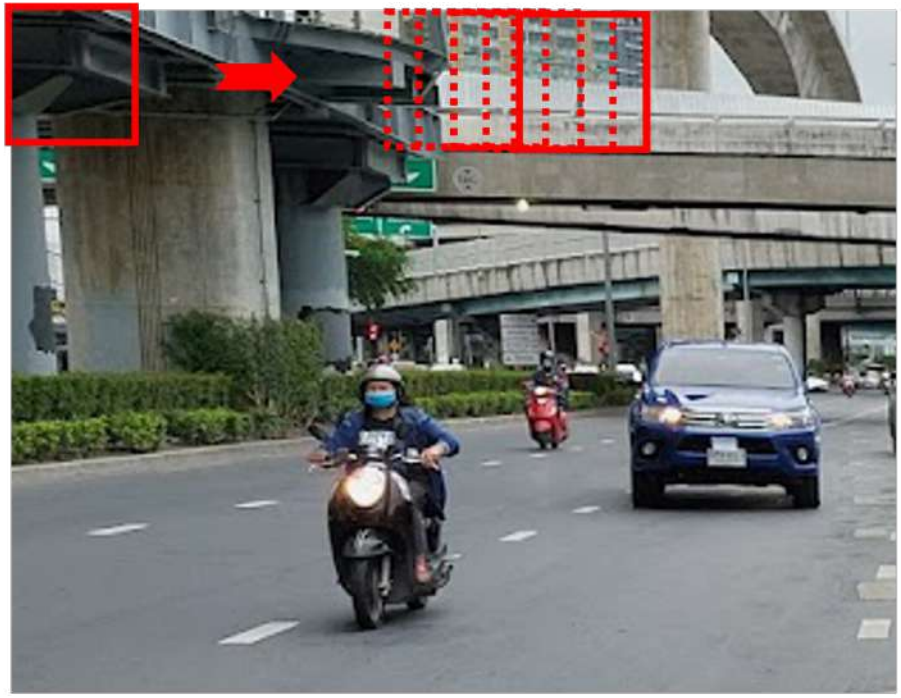
object type

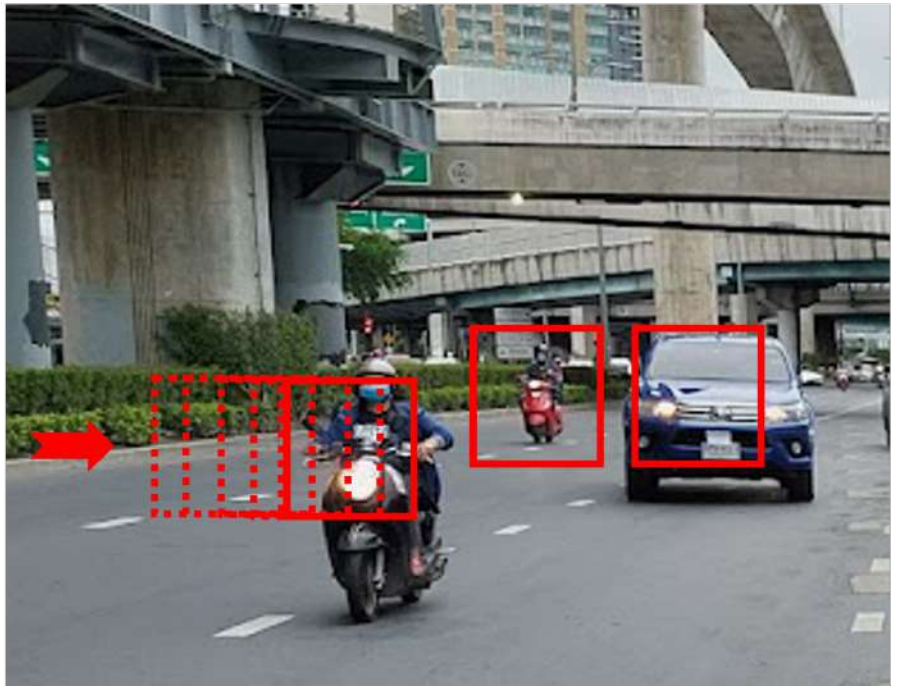
loss function

$$\mathcal{L}(\hat{y}, y) = (\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2 + \ldots + (\hat{y}_8 - y_8)^2, \quad y_1 = 1$$

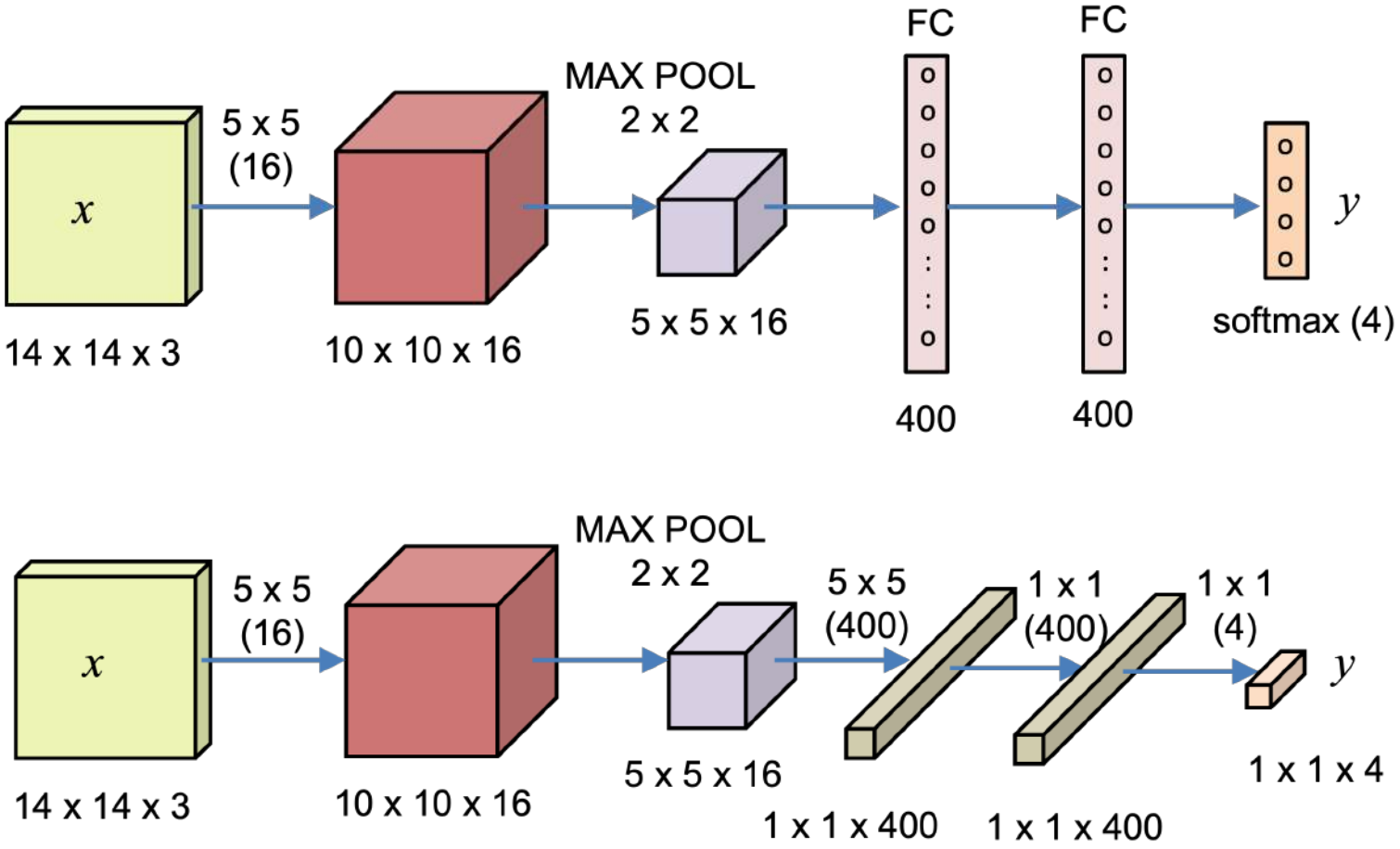
$$\mathcal{L}(\hat{y}, y) = (\hat{y}_1 - y_1)^2, \quad y_1 = 0$$

Object detection with sliding window

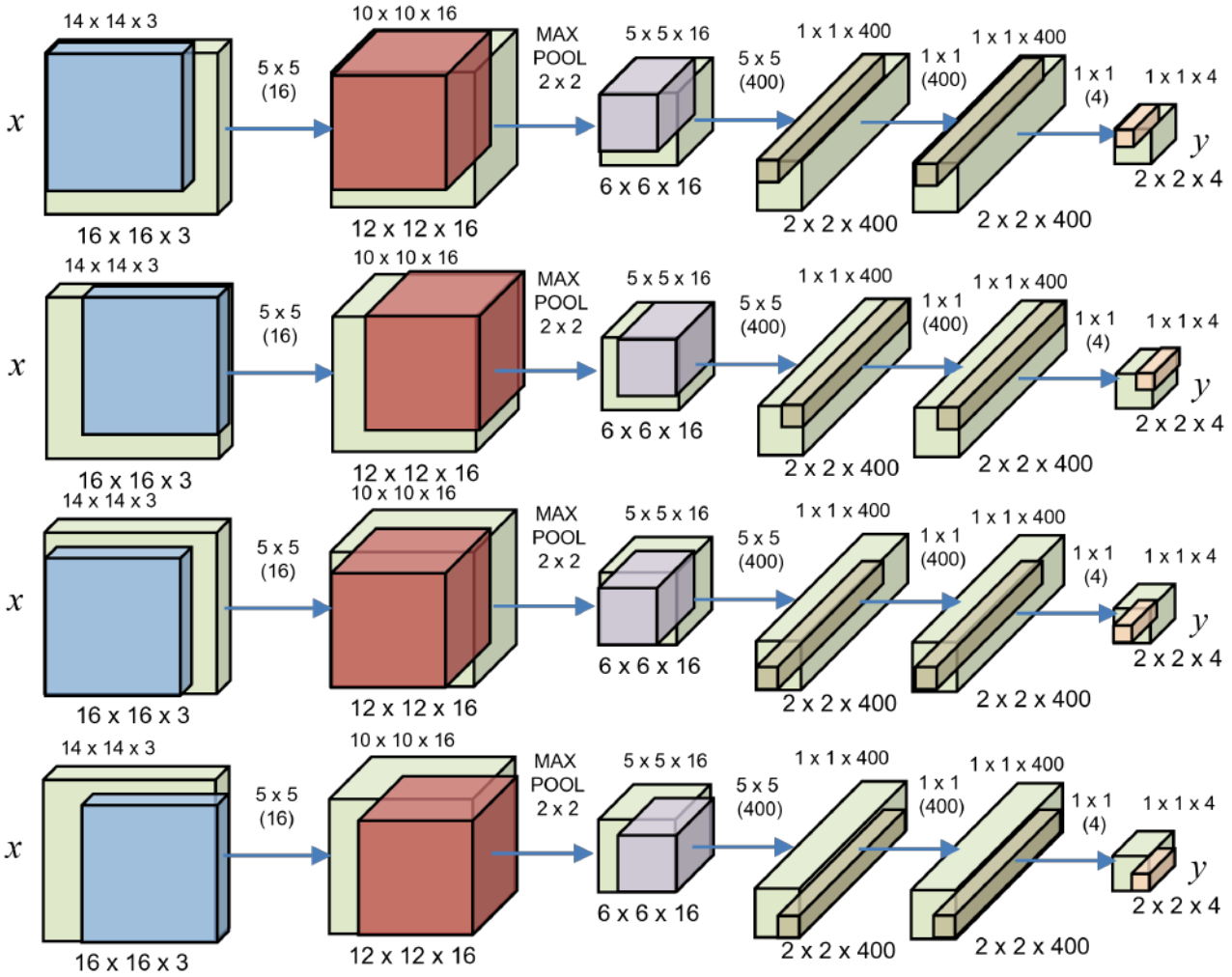




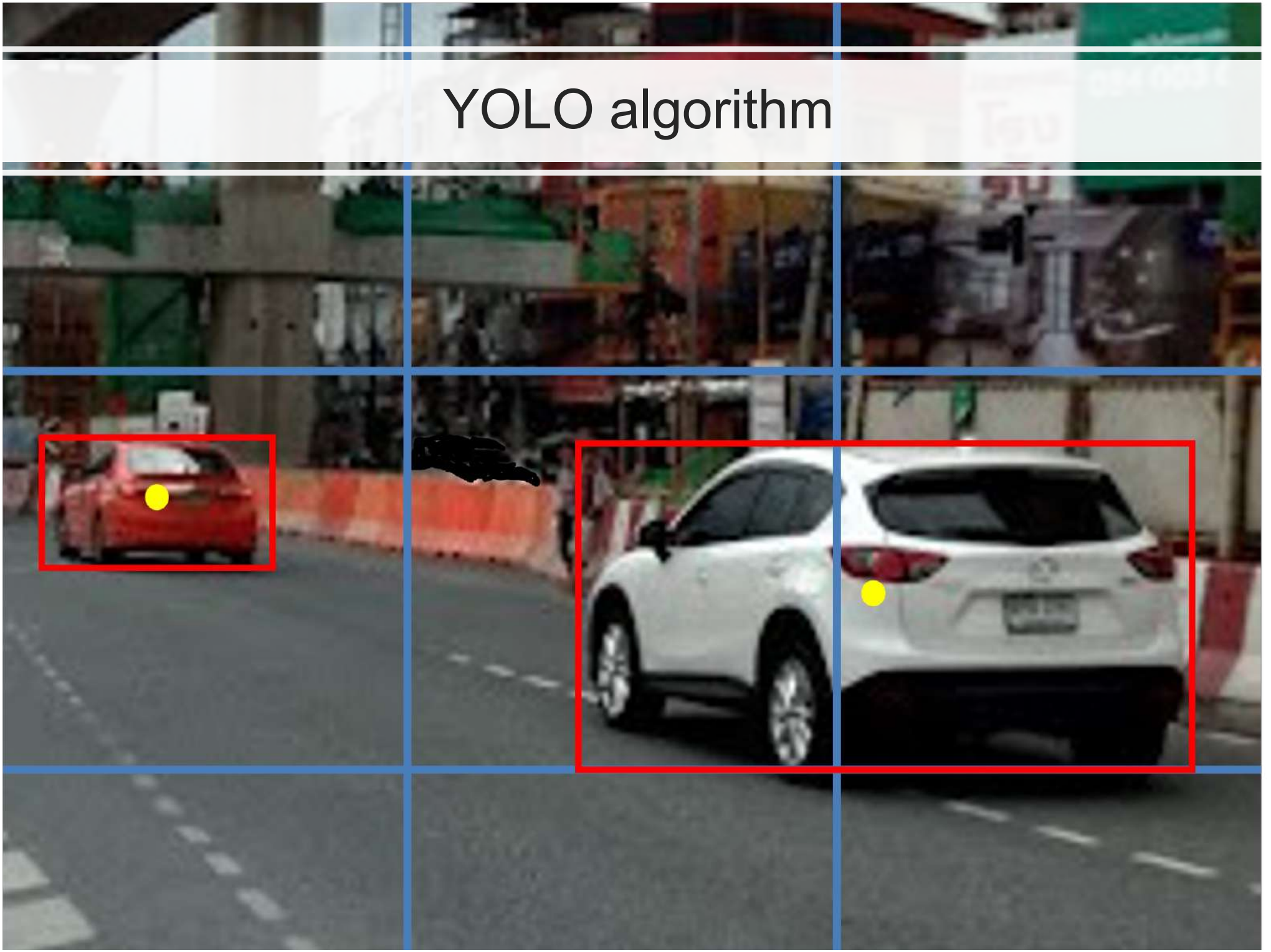
CNN with FC changed to convolutional layers



Sliding window replaced by CNN

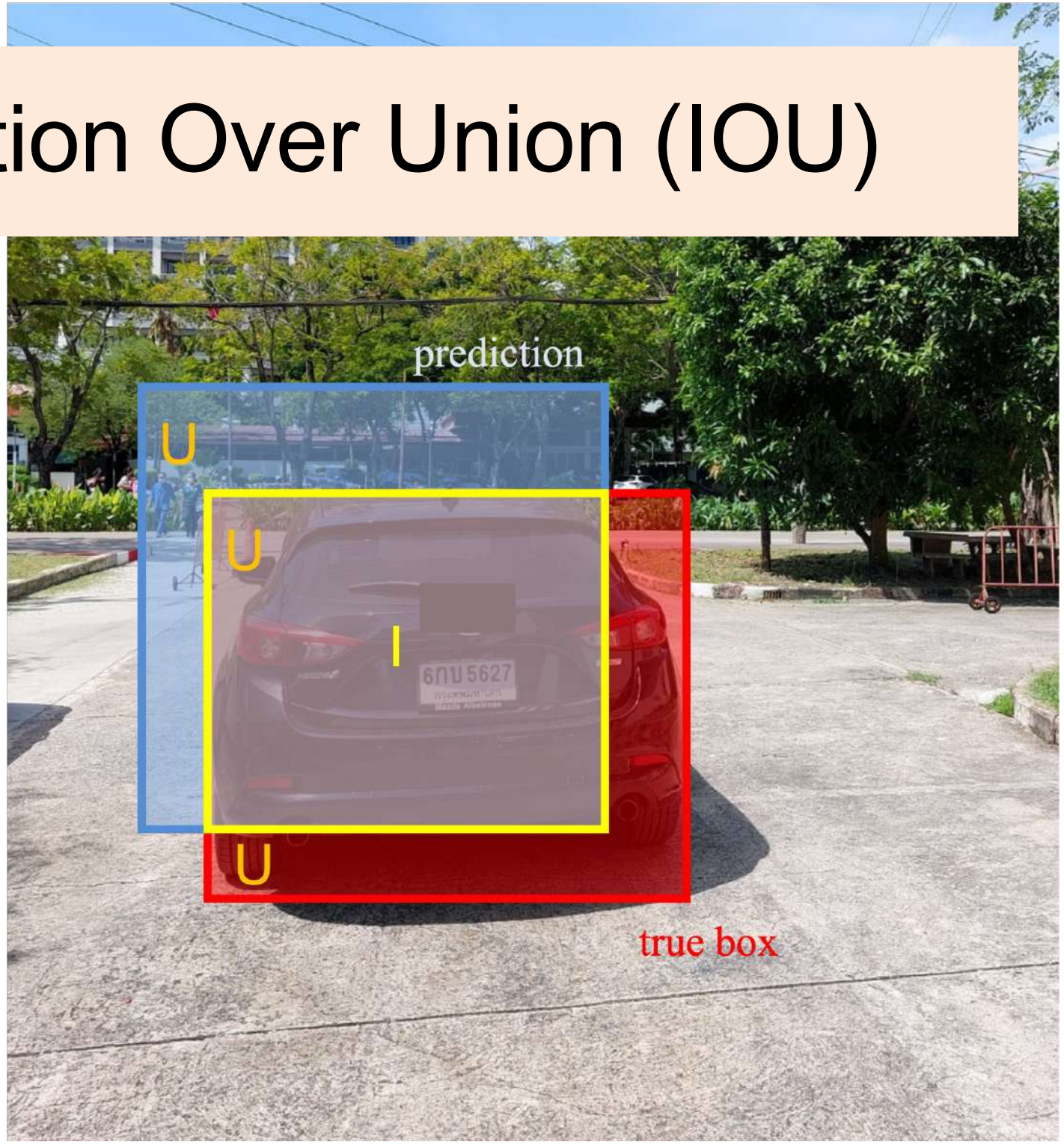


YOLO algorithm



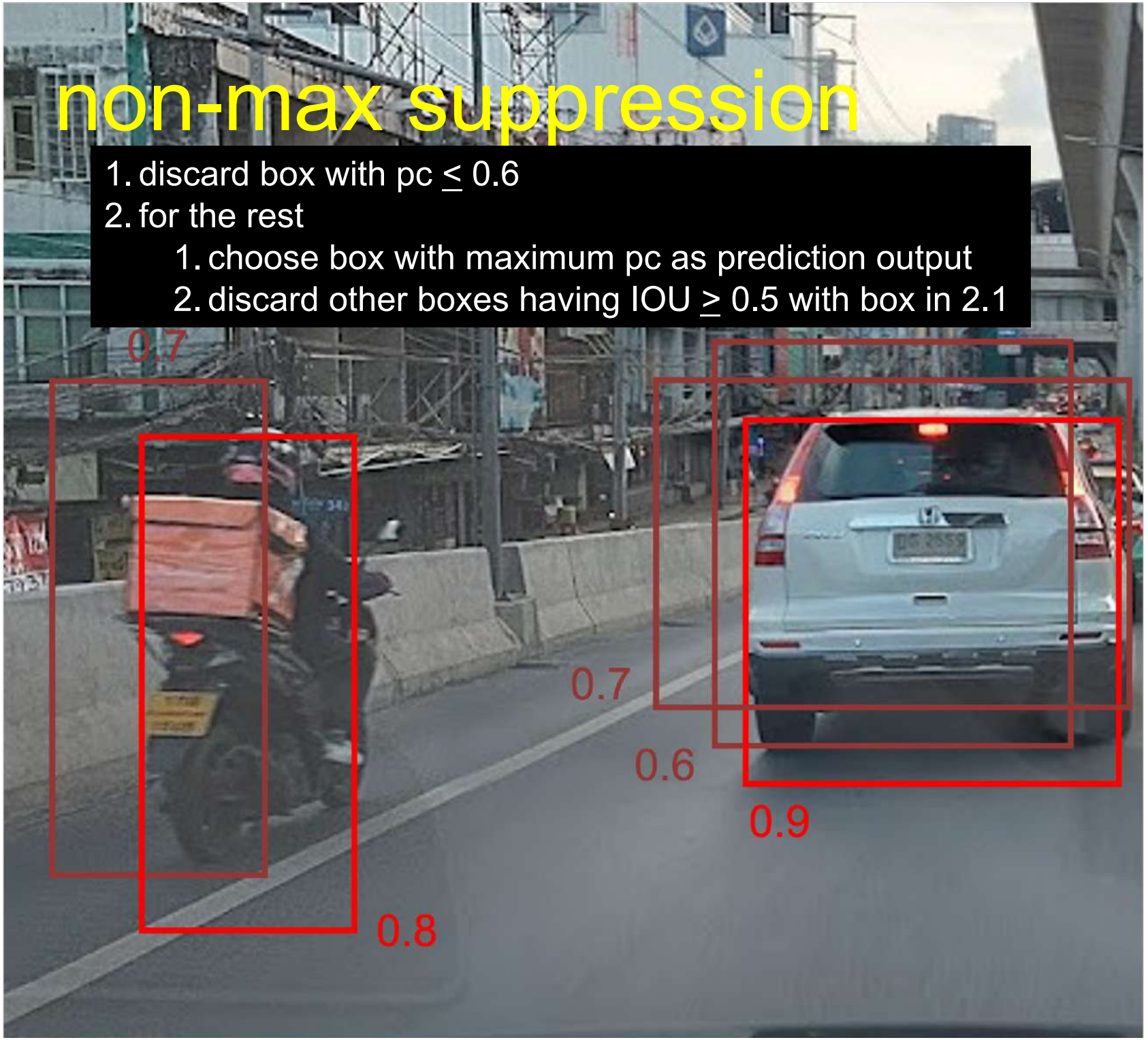
Intersection Over Union (IOU)

$$IOU = \frac{I}{U} \leq 1$$

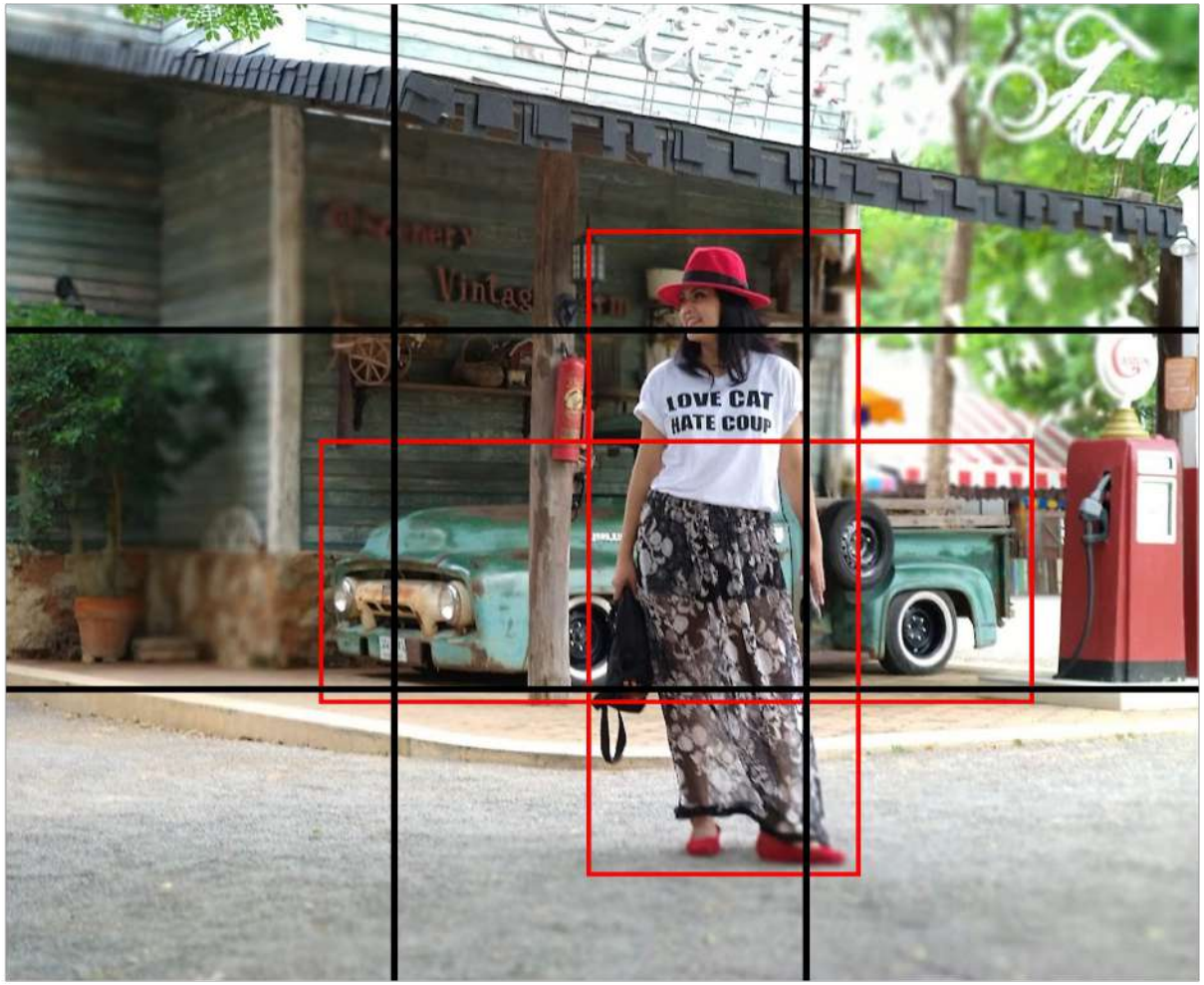


non-max suppression

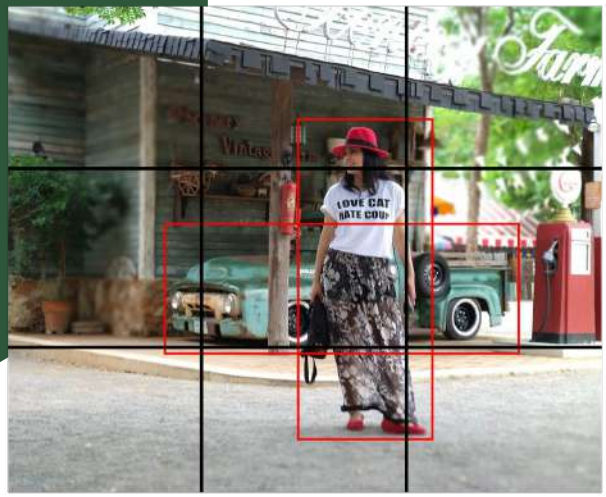
1. discard box with pc ≤ 0.6
2. for the rest
    1. choose box with maximum pc as prediction output
    2. discard other boxes having IOU ≥ 0.5 with box in 2.1



Anchor box

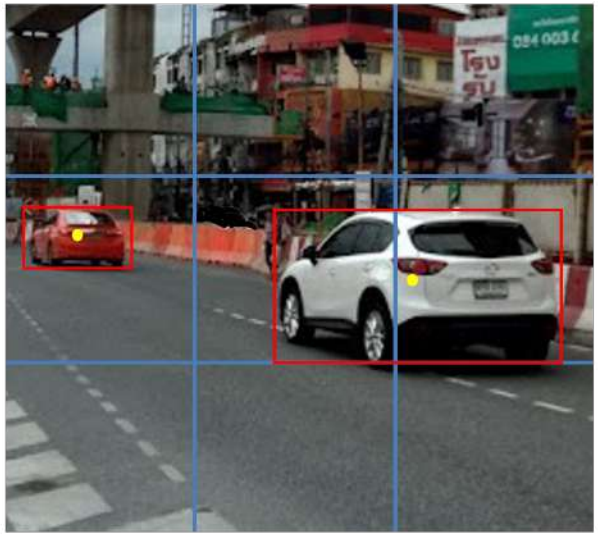


Anchor box output



car and human

$$y = \begin{bmatrix} p_{c1} \\ b_{x1} \\ b_{y1} \\ b_{w1} \\ b_{h1} \\ c_{11} \\ c_{12} \\ c_{13} \\ p_{c2} \\ b_{x2} \\ b_{y2} \\ b_{w2} \\ b_{h2} \\ c_{21} \\ c_{22} \\ c_{23} \end{bmatrix} = \begin{bmatrix} 1 \\ b_{x1} \\ b_{y1} \\ b_{w1} \\ b_{h1} \\ 1 \\ 0 \\ 0 \\ 1 \\ b_{x2} \\ b_{y2} \\ b_{w2} \\ b_{h2} \\ 0 \\ 1 \\ 0 \end{bmatrix}$$



car only

$$y = \begin{bmatrix} p_{c1} \\ b_{x1} \\ b_{y1} \\ b_{w1} \\ b_{h1} \\ c_{11} \\ c_{12} \\ c_{13} \\ p_{c2} \\ b_{x2} \\ b_{y2} \\ b_{w2} \\ b_{h2} \\ c_{21} \\ c_{22} \\ c_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ x \\ x \\ x \\ x \\ x \\ x \\ x \\ 1 \\ b_{x2} \\ b_{y2} \\ b_{w2} \\ b_{h2} \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

don't care

anchor box 1

anchor box 2



Face recognition with CNN

One-shot problem






personnel database

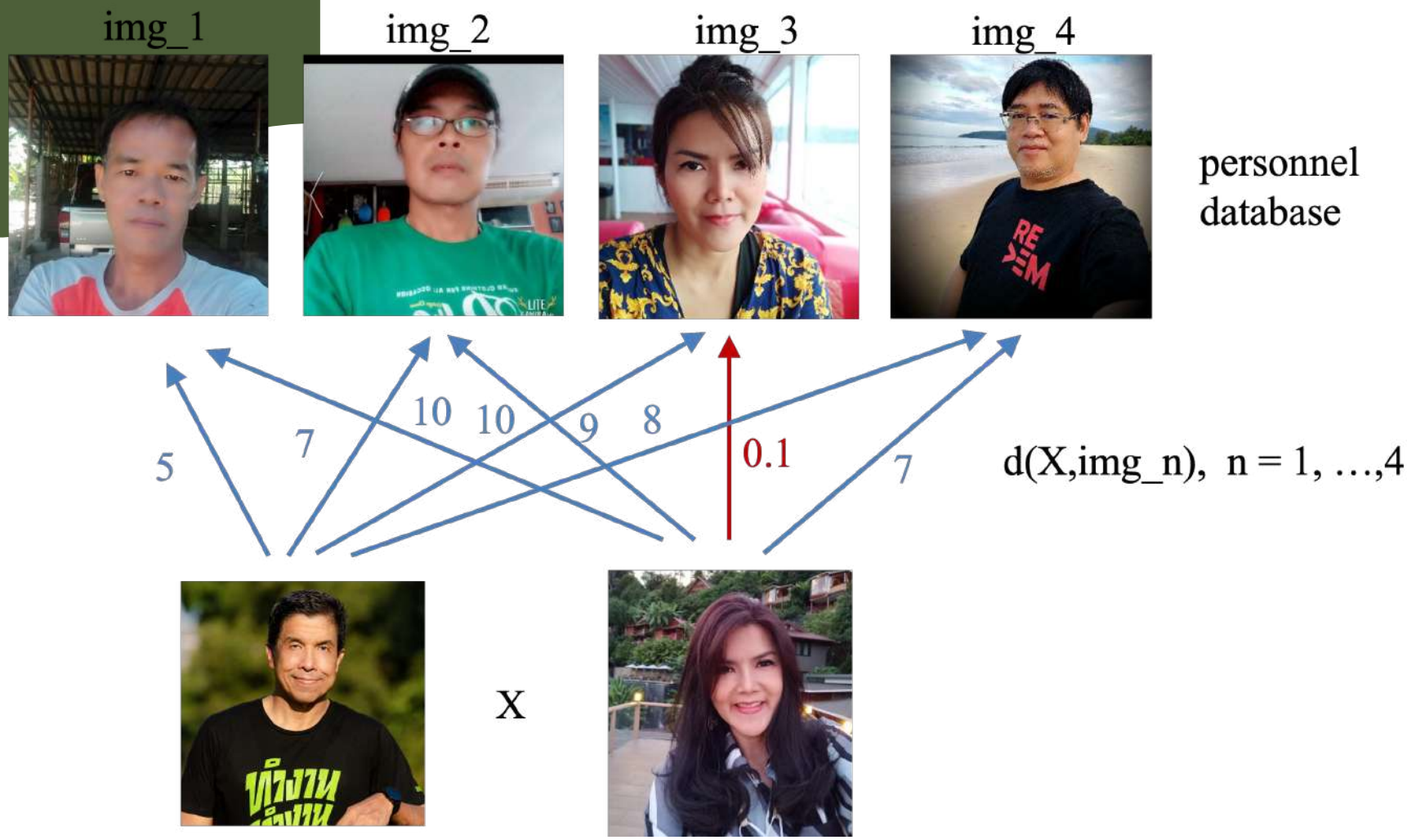
match

?



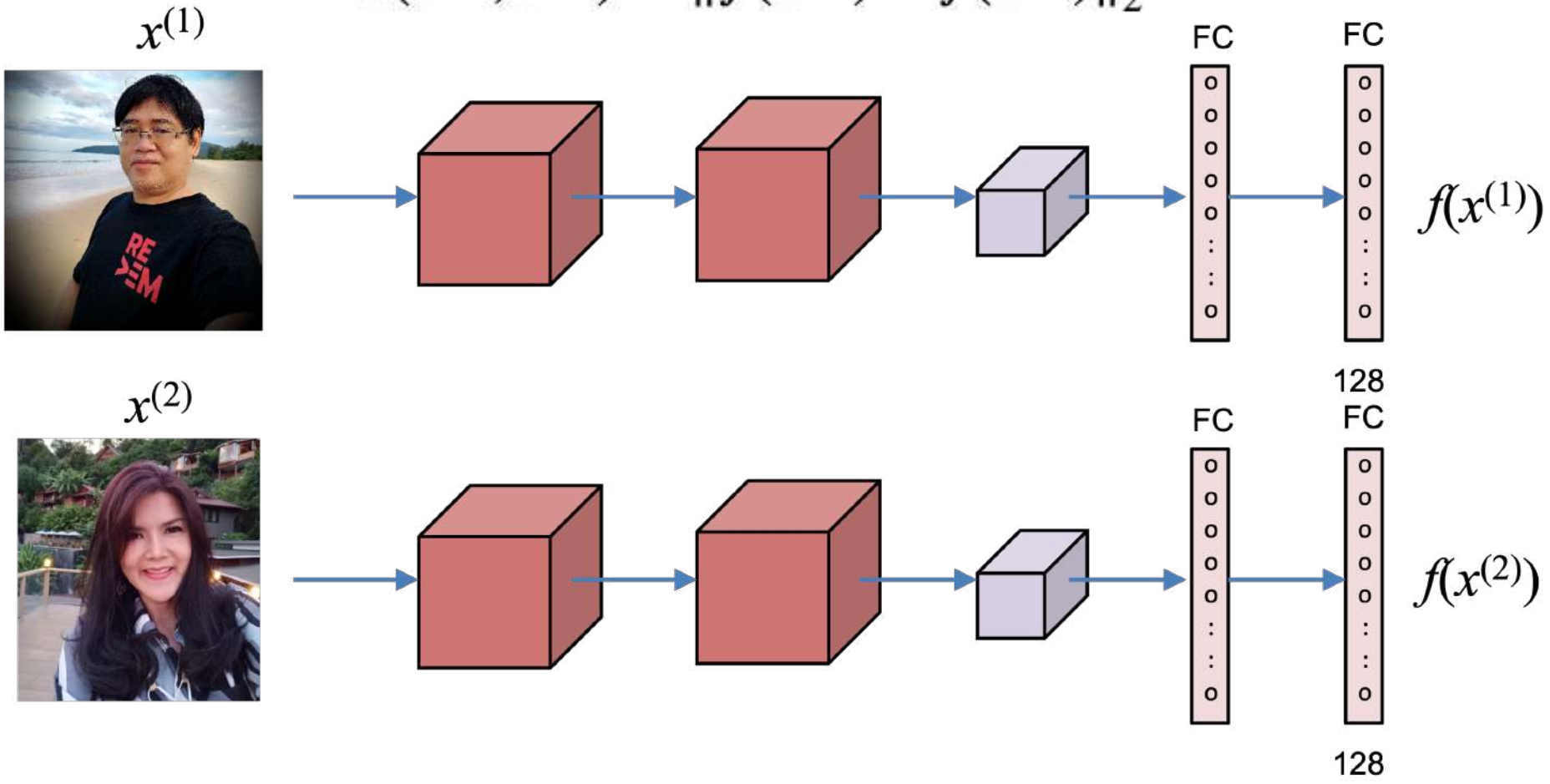

sample images

Similarity function



Siamese network

$$d(x^{(1)}, x^{(2)}) = \|f(x^{(1)}) - f(x^{(2)})\|_2^2$$



Triplet loss



Anchor
(A)



Positive
(P)



Anchor
(A)



Negative
(N)

$$d(A, P) = \|f(A) - f(P)\|_2^2 \leq d(A, N) = \|f(A) - f(N)\|_2^2$$

$$\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 \leq 0$$

$$\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha \leq 0$$

avoid trivial solution

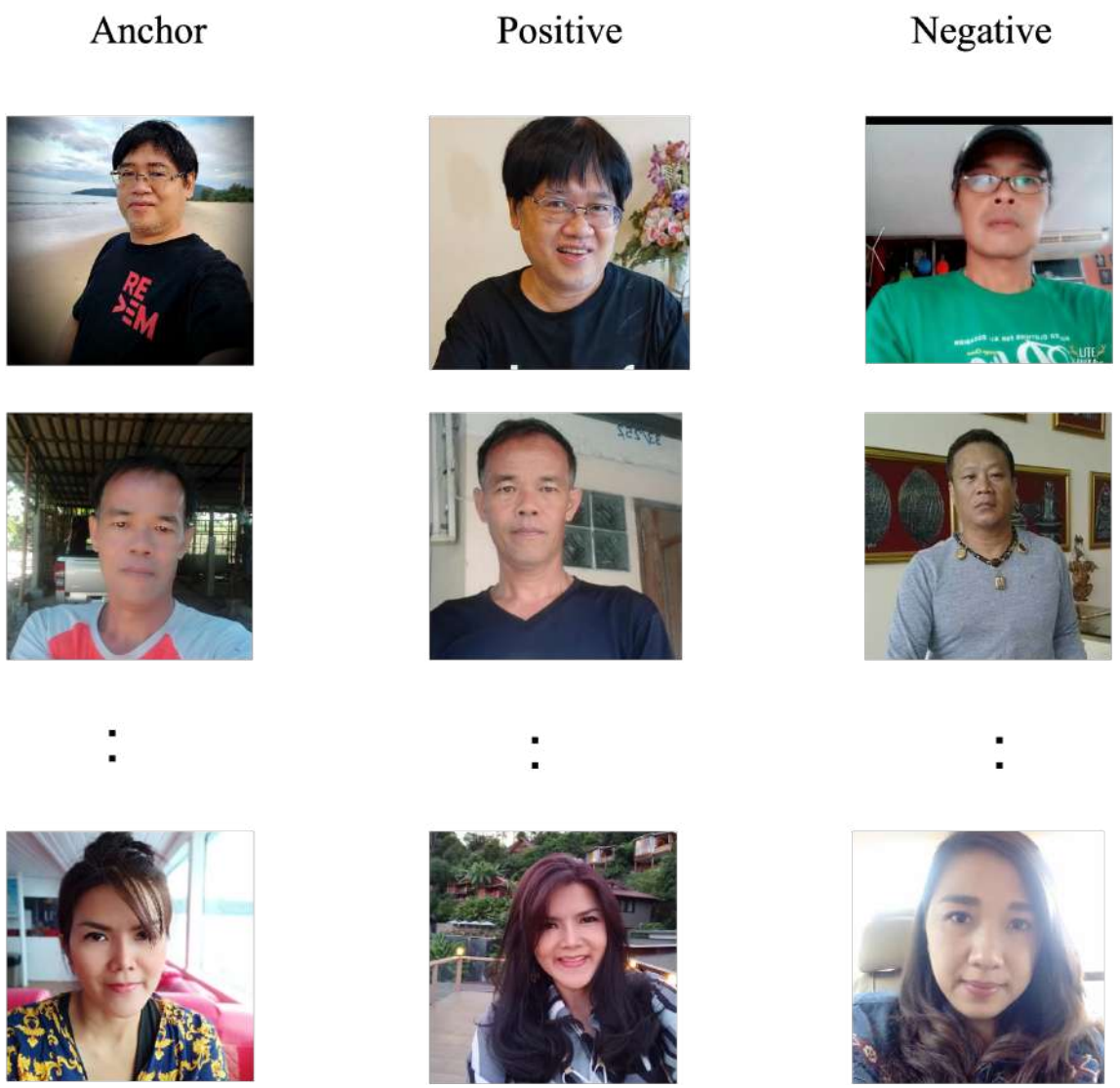
Total cost function

$$\mathcal{J} = \sum_{i=1}^{m} \mathcal{L}(A^{(i)}, P^{(i)}, N^{(i)})$$

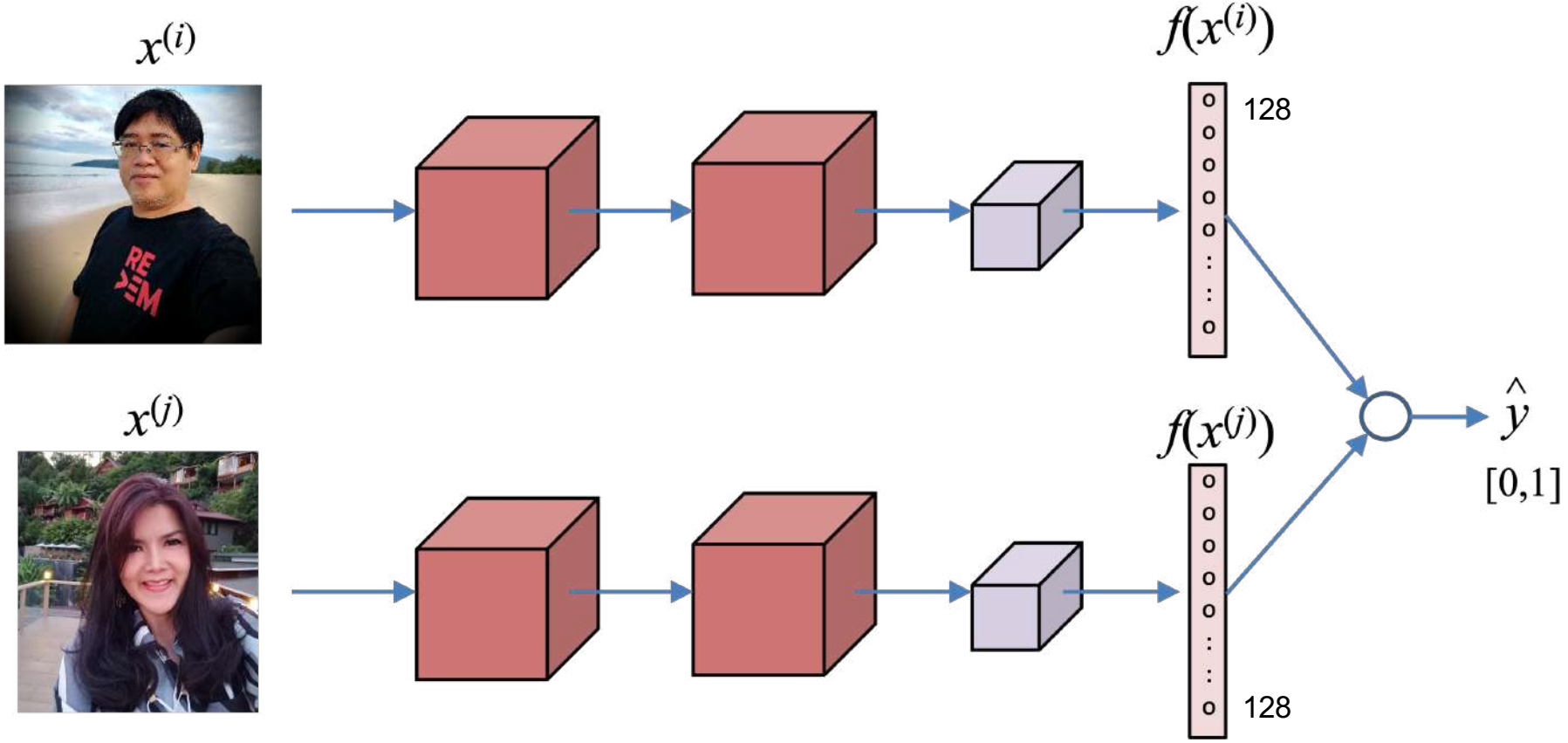
Triplet loss function

$$\mathcal{L}(A, P, N) = max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0)$$

Train model with triplets



similarity function learning



$$\hat{y} = \sigma\left(\sum_{k=1}^{128} w_i |f(x^{(i)})_k - f(x^{(j)})_k| + b_i\right)$$